

**COMSATS University Islamabad, Vehari Campus,
Pakistan**

Department of Computer Science

**Alzheimer's Disease Prediction Using Deep Learning
Techniques**



Software Design Document (SDD)

Submitted in Partial Fulfillment of the Requirements
for the Degree of

Bachelor of Science in Computer Science

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Dedication

We dedicate this work to our parents, whose unwavering support, sacrifices, and prayers have been the foundation of our success. Their encouragement, patience, and belief in our abilities have guided us throughout this journey.

We also dedicate this work to our teachers and mentors, who inspired us to pursue knowledge with dedication and integrity.

Acknowledgement

All praise and gratitude are due for the successful completion of this project. The authors would like to express their sincere appreciation to their supervisor for continuous guidance, valuable suggestions, and constructive feedback throughout the development of this thesis. The supervision and mentorship provided significant support in completing this work successfully.

Special thanks are extended to the faculty members of the Department of Computer Science for their academic support and encouragement during the degree program.

The authors are also deeply grateful to their parents and families for their unwavering support, motivation, and prayers, which have been a constant source of strength.

Finally, appreciation is extended to everyone who directly or indirectly contributed to the successful completion of this project.

Certificate of Approval

This is to certify that the Final Year Project titled “**Alzheimer’s Disease Prediction Using Deep Learning Techniques**” has been developed by **Fatima (FA22-BCS-017)** and **Abu-ul-Hasnat (FA22-BCS-005)** under the supervision of **Dr.Rehan Ashraf** and that in his opinion, the project is adequate in scope and quality for the award of the degree of Bachelor of Science in Computer Science.

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Executive Summary

Alzheimer's disease is a chronic and progressive neurological disorder that primarily affects memory, cognitive abilities, and behavior. It is one of the most common causes of dementia worldwide and presents a growing challenge for modern healthcare systems. With the increasing aging population globally, the number of patients suffering from Alzheimer's disease is also rising rapidly. Early detection of this disease is extremely important as it allows timely treatment, better care planning, and helps slow down its progression. However, conventional diagnostic methods are often expensive, time-consuming, and require expert medical analysis, which can lead to delays in diagnosis. The main problem addressed in this project is the lack of an efficient, accurate, and automated system for early prediction of Alzheimer's disease. Current approaches depend heavily on manual interpretation of medical data such as MRI scans and clinical reports. These methods are not only slow but can also be influenced by human error. In addition, limited access to specialized doctors in many regions makes early diagnosis even more difficult. Therefore, there is a strong need for an intelligent and automated system that can assist in detecting Alzheimer's disease at an early stage. To solve this problem, this project proposes a deep learning-based system for Alzheimer's disease prediction. The system will use advanced machine learning techniques, especially Convolutional Neural Networks (CNNs), to analyze brain MRI images. Deep learning models are capable of automatically learning patterns and extracting important features from complex medical data. The proposed system will classify patients into different stages of Alzheimer's disease, such as mild, moderate, and severe, based on input data. The system includes several key features such as automated data processing, high accuracy prediction, fast results, and a simple user interface. It will also generate visual outputs and reports to help medical professionals better understand the predictions. These features aim to reduce manual workload and improve diagnostic consistency. The technologies used in this project include Python programming language along with deep learning frameworks such as TensorFlow and PyTorch. Data preprocessing techniques will be applied to improve data quality and model performance. Additionally, visualization tools will be used to present results in an understandable format. Model evaluation will be performed using metrics such as accuracy, precision, recall, and F1-score. The expected impact of this project is significant in the healthcare domain. It will provide an intelligent tool for early Alzheimer's disease detection, helping doctors make faster and more accurate decisions. It will also reduce diagnostic errors and improve patient care. This project demonstrates the potential of artificial intelligence in solving real-world medical problems and can be further extended for clinical use in the future. In conclusion, this project bridges the gap between traditional medical diagnosis and modern artificial intelligence techniques by developing a reliable and efficient system for Alzheimer's disease prediction.

Abbreviations

Abbreviation	Full Form
SRS	Software Requirements Specification
SDD	Software Design Document
API	Application Programming Interface
UI	User Interface
DB	Database

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Chapter 1

Software Design

1.1 Introduction to Software Design

The chapter involves the design of the Alzheimer Disease Prediction System and this is detailed in the chapter. It transforms the requirements specified in Chapter 3 into a formal system architecture and design models. The idea is to offer a clear and structured road map to implementation, which will make the system scalable, maintainable and efficient. The system is aimed at analyzing the images of MRI scan with the use of the deep learning methods and determining the stage of the Alzheimer disease. This chapter encompasses architectural design, UML diagrams, data flow and database design that are specifically oriented towards a machine learning-based healthcare system.

1.2 Design Methodology and Software Process Model

1.2.1 Design Methodology

The suggested Alzheimer Disease Prediction System adheres to a hybrid design approach that involves Client-Server Architecture, Model-View-Controller (MVC) and Data-Centric Design. This combination has been chosen in order to adequately handle both the machine learning processing and user interaction aspects of the system. The reason behind the Client-Server Architecture is that the system will be using a frontend interface (client) and a back-end server which will process and communicate with the deep learning model. This configuration will be scalable, and more than one user (e.g., doctors or administrators) can use the system at the same time.

- OOD
- MVC
- Layered Architecture
- Microservices Architecture
- Client-Server Architecture
- Data-Centric Design (for ML systems)

Model-View-Controller (MVC) Pattern:

The Model represents the deep learning component and data processing, including MRI images and prediction results. The View is the user interface where users upload MRI scans and view the prediction results. The Controller handles user input and acts as an intermediary between the frontend and backend. Additionally, a Data-Centric Design methodology is applied since the system is highly dependent on MRI image data and machine learning models. The performance and accuracy of the system depend on the quality, preprocessing, and flow of data. Proper handling of datasets ensures accurate predictions and effective model training.

Rationale of Choice of Approach:

Such a combination of methodologies is well-suited for a machine learning-based healthcare application due to the following reasons:

- Client-Server Architecture allows remote access and efficient processing of large MRI datasets.
- The MVC pattern improves maintainability by separating data, interface, and application logic.
- Data-Centric Design ensures that the system focuses on data quality, preprocessing, and model performance, which are critical in deep learning applications.

Design Principles Followed

The system design is based on standard software engineering principles to ensure high-quality development:

- **SOLID Principles**
Each module is designed with a single responsibility (e.g., preprocessing, prediction, UI handling). The system is structured to allow easy extension without modifying existing code.
- **DRY (Don't Repeat Yourself)**
Reusable functions are implemented for preprocessing and prediction tasks to avoid redundancy.
- **Separation of Concerns**
The frontend, backend, and machine learning model are separated to improve maintainability and clarity.
- **Modularity**
The system is divided into independent modules such as image processing, prediction engine, and database management.
- **Scalability**
The architecture is designed to support future enhancements, including more advanced models and larger datasets.

The selected design methodology ensures that the Alzheimer's Disease Prediction System is efficient, scalable, and maintainable. By combining Client-Server Architecture, MVC, and Data-Centric Design, the system effectively handles both user interaction and deep learning-based prediction tasks.

1.2.2 Software Process Model

The Alzheimer Disease Prediction System proposed is based on the Incremental Software Development Model. The reason this model is chosen is that this system is built around a machine learning pipeline in which various components (data preprocessing, model training, prediction, and user interface development) can be built and refined in distinct stages.

Rationale behind using Incremental Model

Incremental model is the best model to be used in this project because of the following reasons:

- It is an ML-based system, as model improvement occurs over time as a result of training and tuning.
- Each module (data preprocessing, CNN model, frontend, backend) can be developed separately.
- It enables the provision of a simple working system early on with continuous enhancements.
- It facilitates adaptability in updating datasets and model optimization.

This approach allows testing of every component of the system and improves it progressively rather than building everything at once. This way, one can test every component of the system and refines it progressively rather than building it all at once.

Iteration Cycles / Development Phases

The system development is split into several increments:

Increment 1: Data Collection & Preprocessing

- MRI dataset is collected and cleaned.
- Image resizing, normalization, and augmentation are applied.

Increment 2: Model Development

- CNN is developed.
- Training of the model is performed using processed MRI images.
- Preliminary accuracy measures are evaluated.

Increment 3: Backend Integration

- Backend server is integrated with the model.
- APIs are designed for prediction requests.

Increment 4: Frontend Development

- User interface is developed.
- Users can upload MRI scans and view results.

Increment 5: System Integration & Optimization

- All modules are integrated into a single system.
- Performance tuning and optimization of model and system is performed.

Testing and Validation

Incremental testing is included at each stage to ensure system reliability:

- **Unit Testing:** Each module (preprocessing, model, backend) is tested individually.
- **Integration Testing:** Ensures smooth communication between frontend, backend, and ML model.
- **Model Validation:** CNN model is evaluated using metrics such as accuracy, precision, recall, and loss.
- **User Acceptance Testing (UAT):** The system is tested with sample users (e.g., doctors or testers) to verify usability and prediction accuracy.

Continuous testing throughout development ensures early detection of errors and improves overall system performance.

1.3 System Overview

The Alzheimer Disease Prediction System is a healthcare application based on machine learning that is aimed at helping to detect and identify the stages of Alzheimer disease early. The system communicates with various external organizations such as patients, doctors, administrators and third party services like cloud storage and notification systems.

External Actors:

- Patients End Users: Enter personal details, medical history and access prediction results.
- Doctors / Clinicians: Discuss patient data, make predictions, and interpret reports.
- Administrators: Control users, datasets, system configurations and system performance.
- Third-Party Services: Include cloud storage, email/SMS notification services, and outside APIs.

Major Subsystems:

- item User Interface (Web/Mobile Application)
- item Data Management Module (Stores and processes patient data)
- Machine Learning Prediction Engine (Predicts the stage of Alzheimer disease)
- Report Generation Module (Produces prediction reports to the users and doctors)
- Authentication and Authorization Module (Controls access and secure login)
- Notification Service (Sends alerts and updates through email/SMS)

Input to System:

- demographics (age, gender, etc.)
- Medical history and clinical records
- Cognitive test results (e.g., memory tests, MMSE scores)
- MRI or brain imaging data (supply, when possible)
- Lifestyle information (sleep, diet, exercise)
- Lab test results (blood biomarkers, etc.)

External System Interactions:

- item Cloud Storage: Stores datasets, models and generated reports.
- Email/ SMS services: Dispatches notifications, alerts and results to users.
- Medical Data Sources: Hospitals and laboratories that offer clinical data.

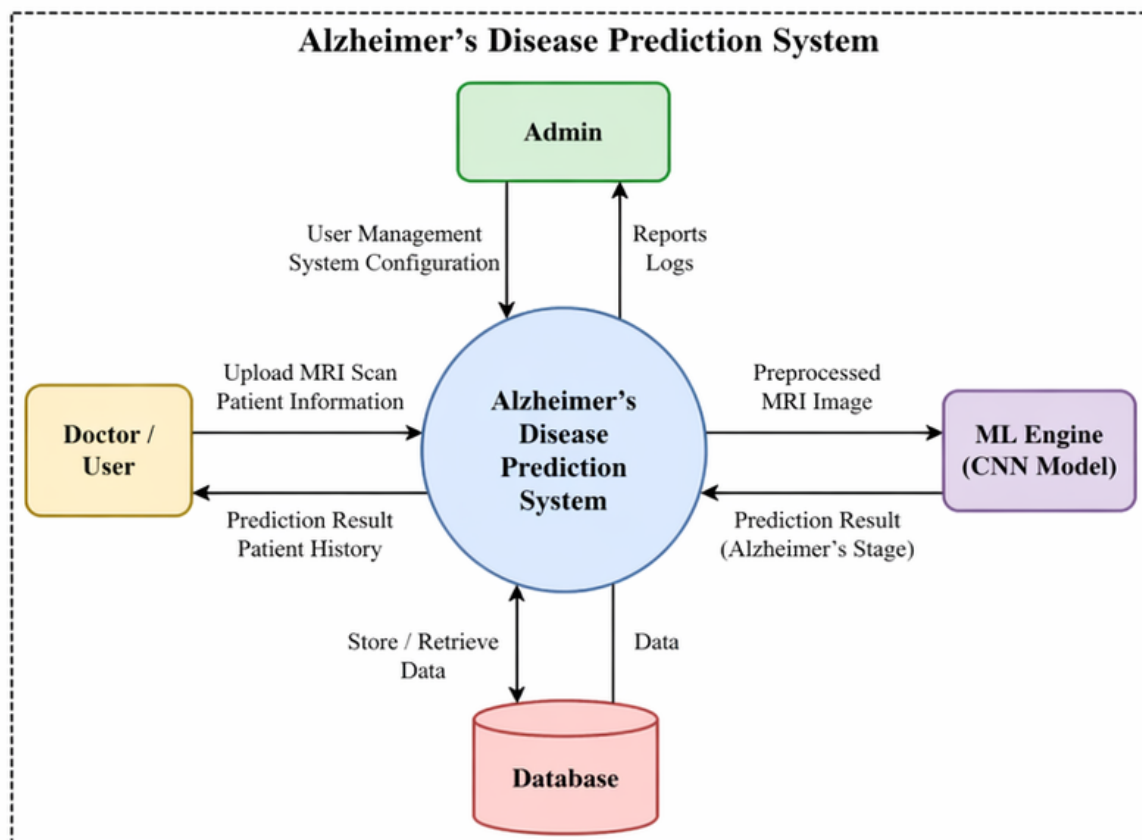


Figure 1.1: Sample Context Diagram of Proposed System

1.4 Architectural Design

A layered architecture is used to design the system in order to maintain modularity, scalability, and easy maintenance. It is further split into three major layers: Presentation Layer, Business Logic Layer, and Data Layer.

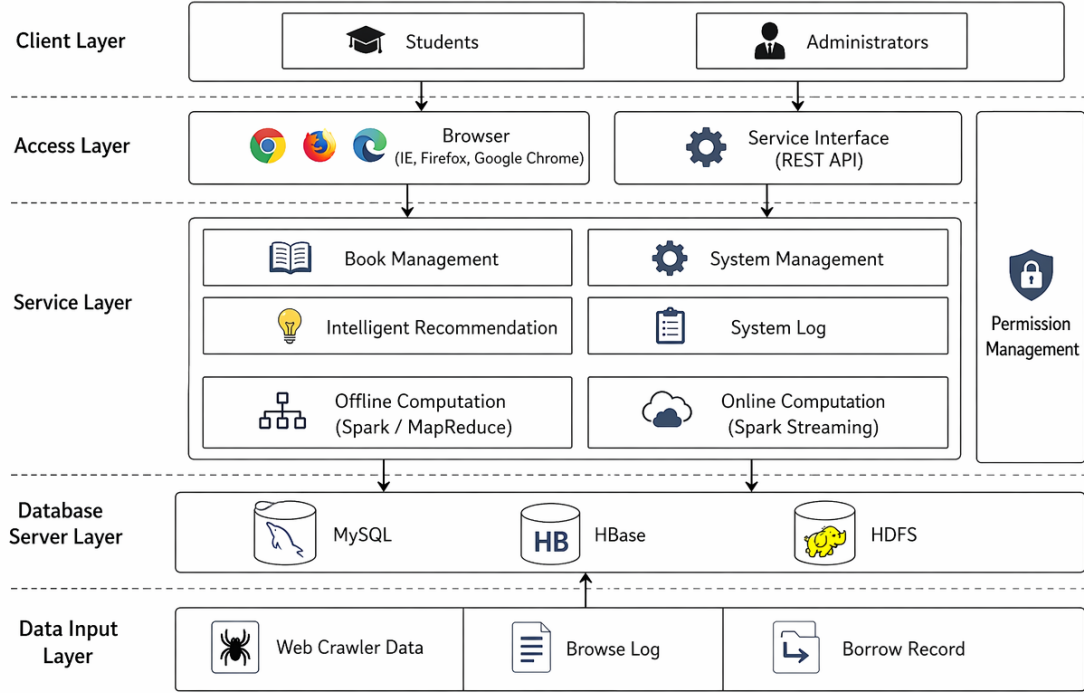


Figure 1.2: Sample System Architecture Diagram of Proposed System

1.4.1 Presentation Layer (Frontend)

This layer deals with the interaction of the system with the users.

- Provides an easy-to-use interface for uploading MRI scans.
- Displays prediction outcomes (Normal, Mild, Moderate, Severe Alzheimer stage).
- Shows graphical reports and analysis results.
- Developed using web or mobile technologies (React, HTML/CSS, Flutter).

1.4.2 Business Logic Layer (Backend + ML Processing)

This is the central layer where all processing and decision-making takes place.

Data Preprocessing Module

- Receives MRI images from the frontend.
- Performs resizing, normalization, and noise removal.
- Converts data into model-compatible format.

Model Training Module

- Uses processed MRI dataset.
- Trains deep learning models (CNN-based architecture).
- Saves trained model for future predictions.
- Optimizes model using hyperparameter tuning.

Prediction Engine

- Loads trained model.
- Takes new MRI scans as input.
- Predicts Alzheimer stage based on patterns.
- Returns results to frontend.

Evaluation Module

- Evaluates model using accuracy, precision, recall, and F1-score.
- Improves model performance.
- Used during training and testing.

API Layer

- Connects frontend and backend.
- Handles requests and responses.
- Transfers MRI data and prediction results.

1.4.3 Data Layer (Database)

This layer manages all system data.

- Stores MRI datasets used for training.
- Saves user-uploaded scans if required.
- Stores prediction history and results.
- Uses databases like MySQL, MongoDB, or Firebase.

1.4.4 Interaction Between Components

1. User uploads MRI scan through Presentation Layer.
2. Data is sent to backend via API Layer.
3. Preprocessing Module cleans and prepares the image.
4. Prediction Engine processes the image using trained model.
5. Model predicts Alzheimer stage.
6. Result is sent back through API Layer.
7. Frontend displays prediction to user.
8. Data is optionally stored in database for future use.

1.5 Design Models

The proposed system design is represented using the following UML diagrams:

- **Use Case Diagram:** Illustrates the interaction between the user and the system, showing the main functionalities of the Alzheimer's prediction system.
- **Class Diagram:** Describes the static structure of the system, including key classes such as User, MRI Scan, Preprocessing Module, CNN Model, and Database along with their relationships.
- **Sequence Diagram:** Shows the step-by-step interaction between system components from MRI upload to final prediction result generation.
- **Activity Diagram:** Represents the workflow of the system, including preprocessing, model prediction, decision-making, and result storage.

These UML diagrams collectively provide a clear understanding of the system architecture, workflow, and component interactions before implementation.

1.5.1 Activity Diagrams

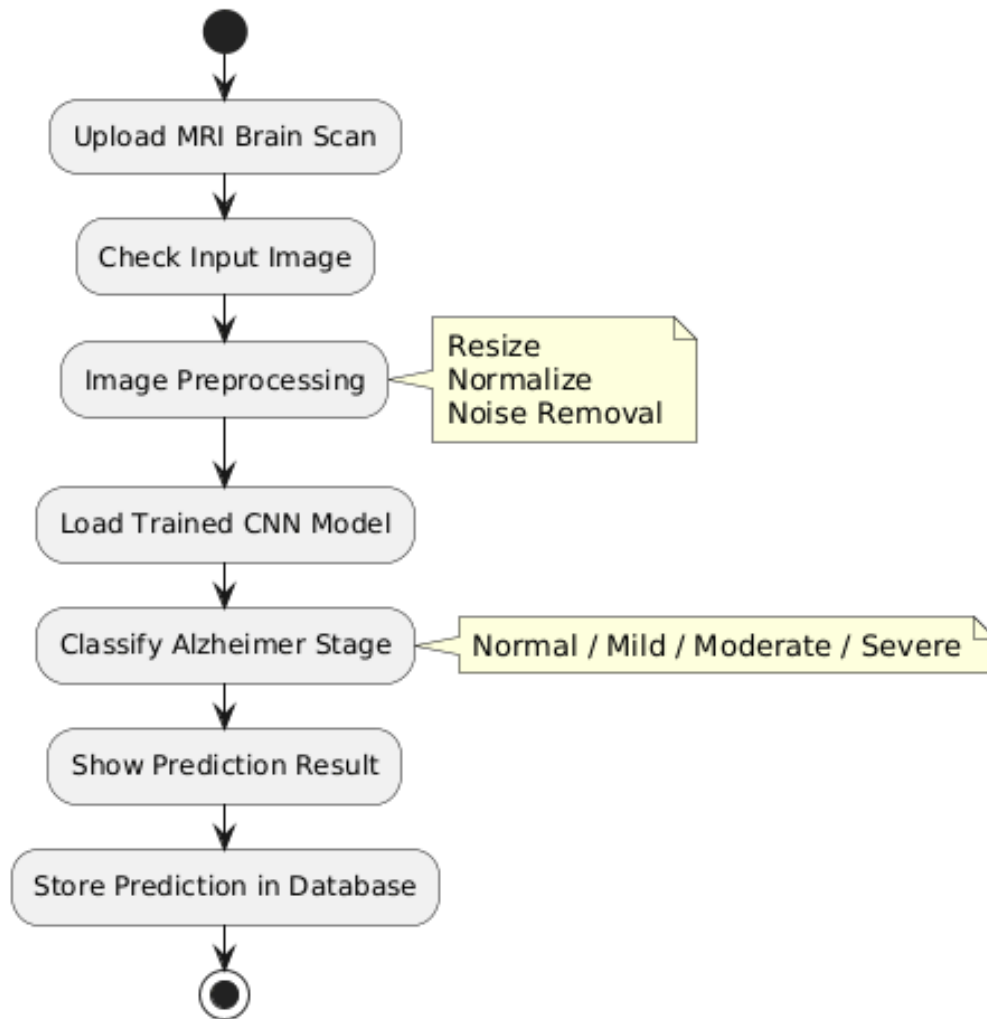


Figure 1.3: Activity Diagram

Module 1: User Management

Description: This activity diagram is a representation of the user-related workflow of the Alzheimer's Disease Prediction System. It comprises of user registration, user login authentication and access control. MRI scans can be uploaded and prediction results analyzed only by authorized users (doctor/patient).

Main Activities:

- User Registration
- User Login
- Authentication Verification item Access System Dashboard.

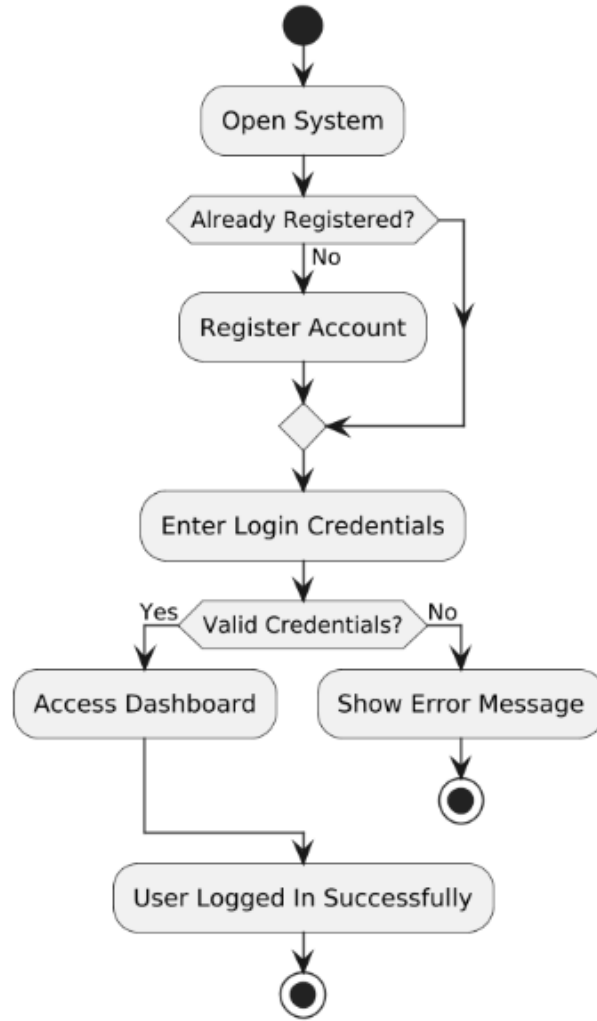


Figure 1.4: Activity Diagram – User Management Module

Module 2: Alzheimer Prediction Module

Description: The following activity diagram represents the fundamental operation of the Alzheimer’s Disease prediction system based on deep learning. It demonstrates how MRI brain scan data undergoes several processing steps such as preprocessing, CNN model prediction, and result generation.

Main Activities:

- Upload MRI Brain Scan
- Check Input Image
- Image Preprocessing (Resizing, Normalization, Noise Removal)
- Load Trained CNN Model
- Classify Alzheimer’s Stage (Normal, Mild, Moderate, Severe)
- Display Prediction Result
- Store Prediction in Database

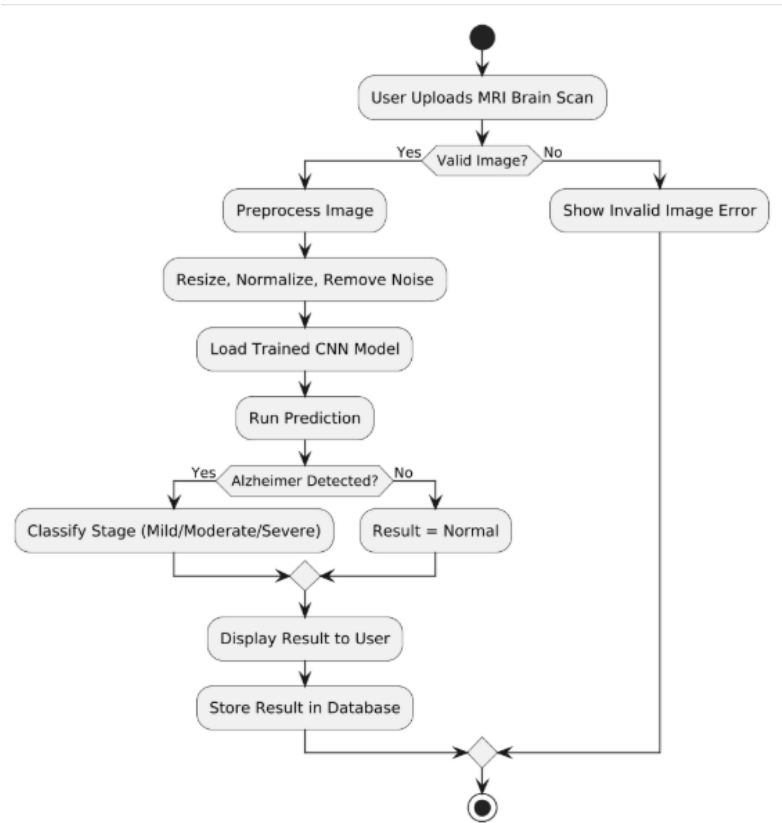


Figure 1.5: Activity Diagram – Alzheimer Prediction Module

Module 3: Admin Module

Description: This module represents the administrative operations of the Alzheimer’s Disease Prediction System. The admin is responsible for monitoring system activity, managing users, viewing prediction reports, and controlling system data.

Main Activities:

- Monitor system activity
- Manage user accounts
- View prediction reports
- Manage MRI data
- System control and maintenance

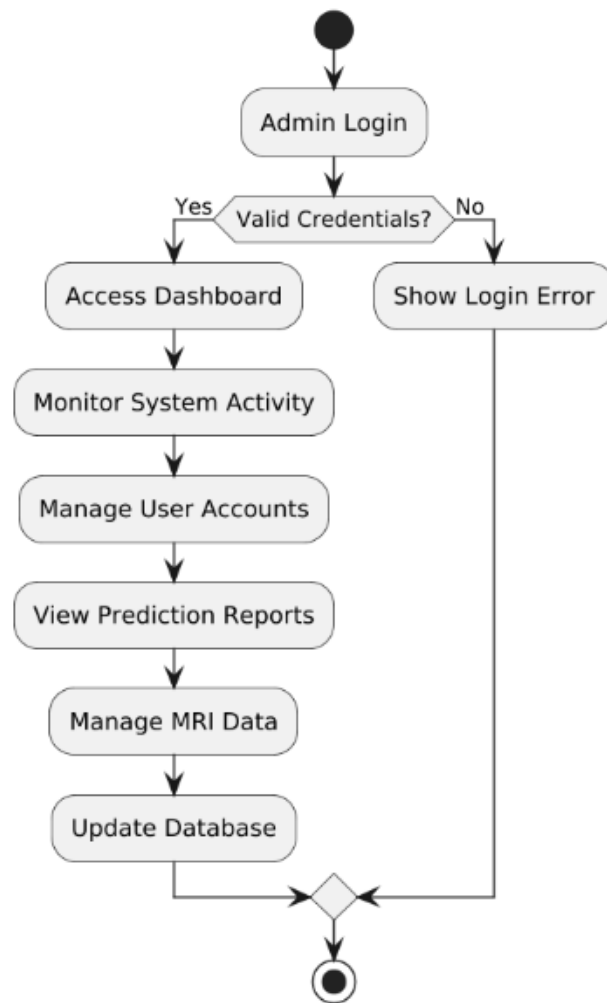


Figure 1.6: Activity Diagram – Admin Module

1.5.2 Class Diagram

The class diagram shows the static structure of the system. The class diagram represents the structure of the system by showing classes, attributes, and relationships. Each class contains specific attributes and methods to perform its role in the system.

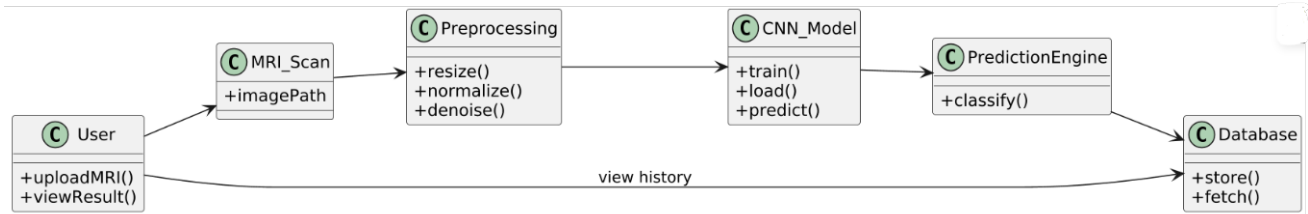


Figure 1.7: class diagram Diagram of Alzheimer's Disease Prediction System

The class diagram of the Alzheimer's Disease Prediction System represents the static structure of the system, including its main classes, relationships, and responsibilities. It provides a clear understanding of how different components interact to achieve MRI-based Alzheimer's prediction using a deep learning model.

Key Classes:

- **User:** Represents the doctor or patient who interacts with the system. The user uploads MRI scans and views prediction results.
- **MRI_Scan:** Represents the input brain MRI image that is processed by the system.
- **Preprocessing Module:** Responsible for image preparation, including resizing, normalization, and noise removal to convert raw MRI data into model-compatible input.
- **CNN Model:** The core deep learning model trained on MRI datasets for feature extraction and classification of Alzheimer's disease stages.
- **Prediction Engine:** Uses the trained CNN model to generate final predictions (Normal, Mild, Moderate, Severe).
- **Database:** Stores user data, MRI scans, and prediction history for future reference.

Relationships:

- The User interacts with the system by uploading MRI scans.
- MRI_Scan is passed to the Preprocessing Module for cleaning and transformation.
- The Preprocessing Module sends processed data to the CNN Model.
- The CNN Model provides output to the Prediction Engine for classification.
- The Prediction Engine stores results in the Database.
- The User can retrieve previous results from the Database.

Responsibilities:

- Ensuring accurate preprocessing of MRI images for reliable prediction.
- Training and managing the deep learning CNN model.
- Generating accurate Alzheimer's disease stage predictions.
- Maintaining prediction history and user data securely.

1.5.3 Sequence Diagram User Login

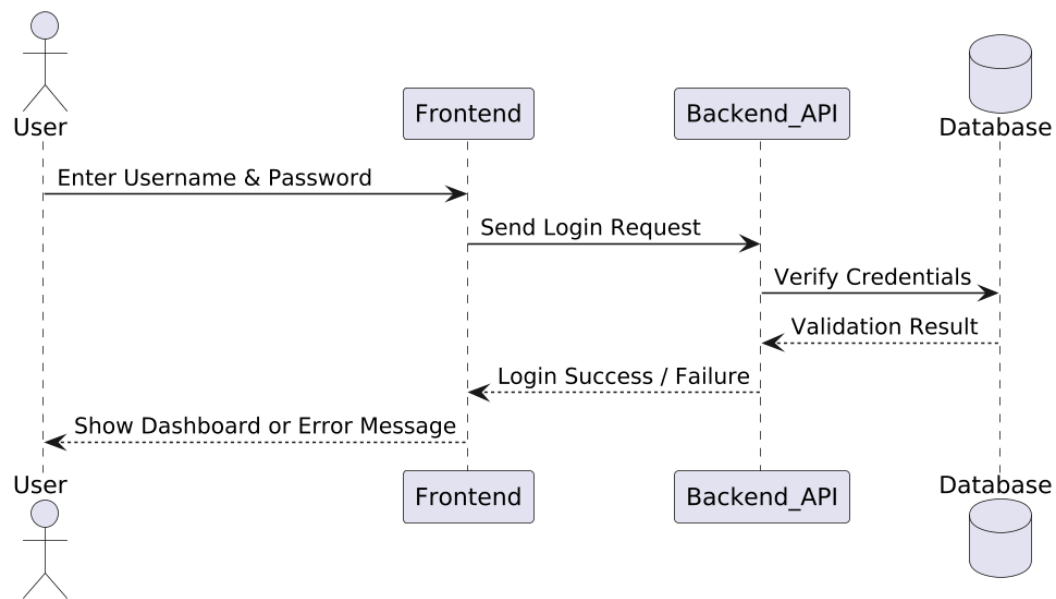


Figure 1.8: Sequence Diagram – User Login Process

1.5.4 Sequence Diagram Core Operation

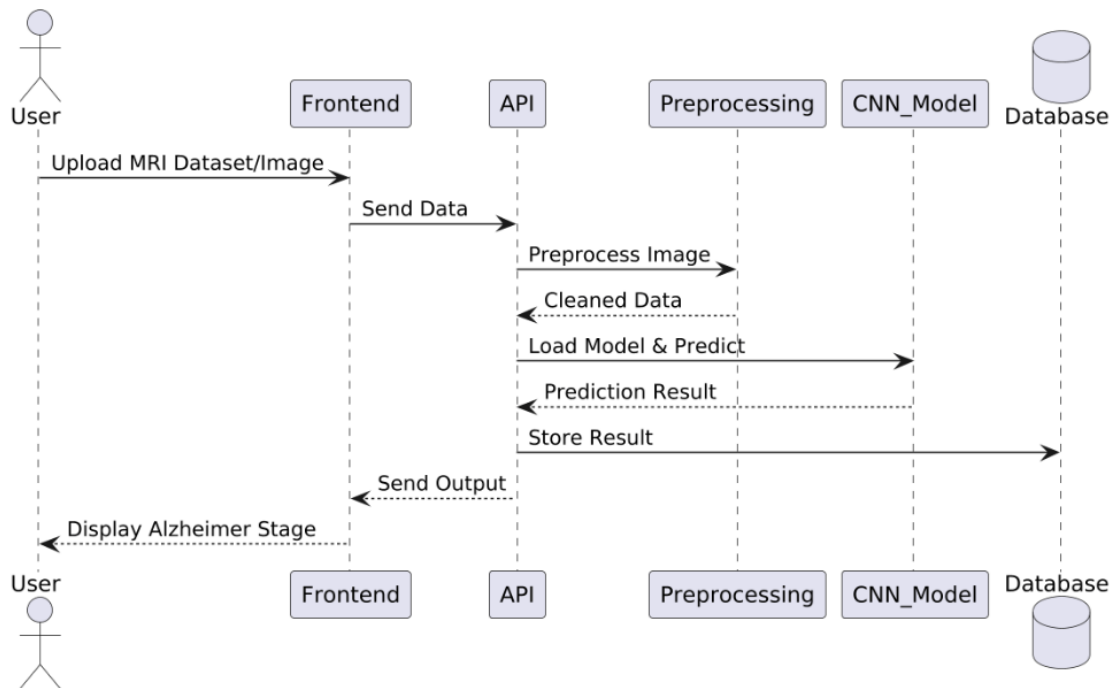


Figure 1.9: Sequence Diagram – Core Prediction Process

1.5.5 State Transition Diagrams

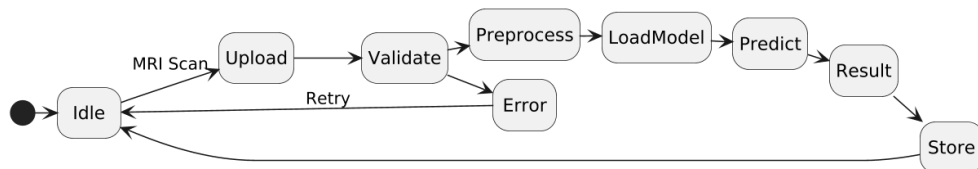


Figure 1.10: State Transition Diagram – Alzheimer Prediction System

1.6 Data Flow Diagrams

1.6.1 Level 1 Data Flow Diagram

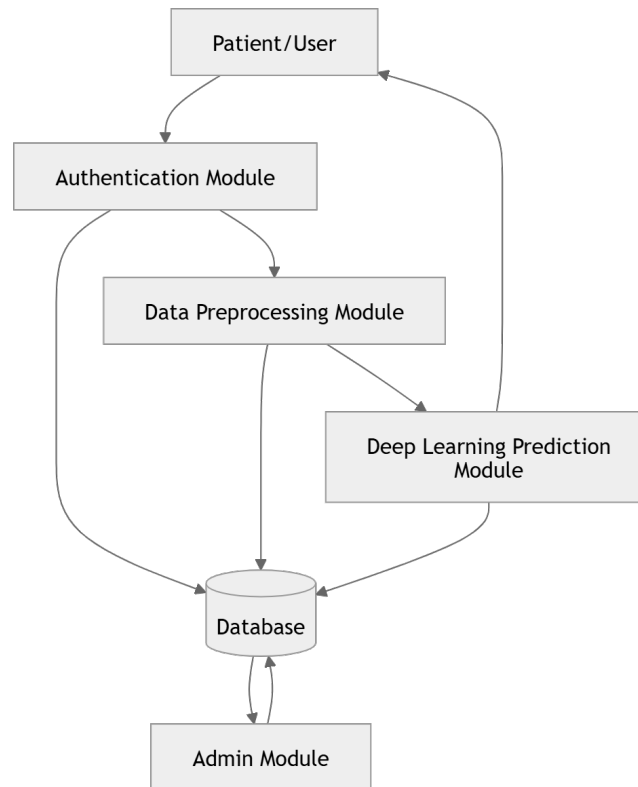


Figure 1.11: Level 1 Data Flow Diagram (DFD-1) of Alzheimer Prediction System

1.6.2 Level 2 Data Flow Diagram

Description: Level 2 DFD further decomposes the internal processing of the Alzheimer's Disease Prediction System into detailed steps including data upload, preprocessing, feature extraction, model training, and prediction generation.

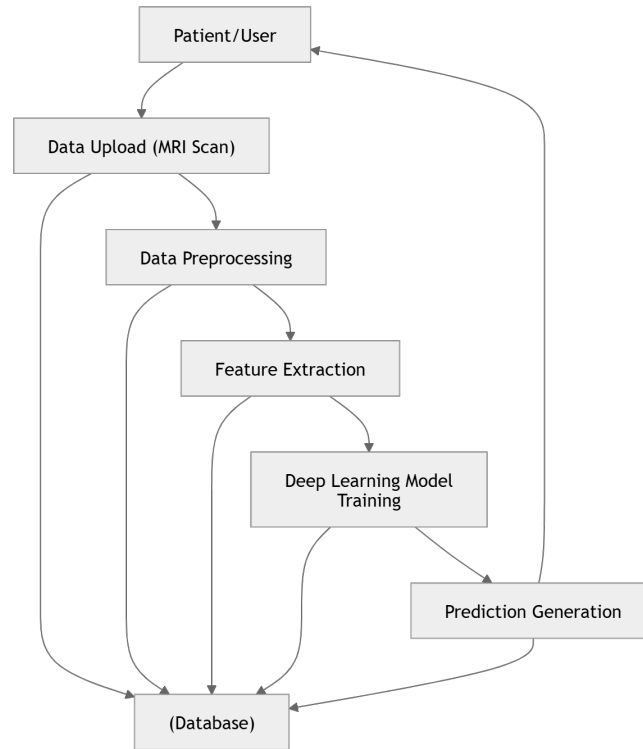


Figure 1.12: Level 2 Data Flow Diagram (DFD-2) of Alzheimer Prediction System

Processes Included:

- Data Upload (MRI Scan Input)
- Data Preprocessing
- Feature Extraction
- Model Training (Deep Learning)
- Prediction Generation

1.7 Data Design

This section describes how data is structured, stored, and managed in the Alzheimer's Disease Prediction System.

Database Type: The system uses a relational database management system (RDBMS) such as MySQL or PostgreSQL. This ensures structured storage, data integrity, and efficient querying of patient and prediction data.

Schema Structure: The database is organized into multiple tables to manage different types of data:

- **Users Table:** Stores user credentials and roles (e.g., user ID, name, email, password).
- **Patient Data Table:** Contains patient-related information and MRI scan references.
- **Prediction Table:** Stores results generated by the deep learning model, including predicted Alzheimer stage and confidence score.

- **Admin Table:** Maintains admin details for system management and monitoring.

Indexing Strategy: To improve performance and ensure fast data retrieval, indexing is applied on key attributes such as user ID, patient ID, and prediction ID. This helps in efficient searching and quick access to records.

Security Mechanisms: The system ensures data security through multiple techniques:

- Password hashing for secure authentication
- Role-based access control (admin and user roles)
- Secure storage of sensitive medical data
- Protection against unauthorized access

Scalability Considerations: The system is designed to handle increasing amounts of data efficiently. It can be scaled by:

- Using cloud-based storage solutions
- Optimizing database queries
- Supporting distributed data processing for large datasets

1.7.1 Entity Relationship Diagram (ERD)

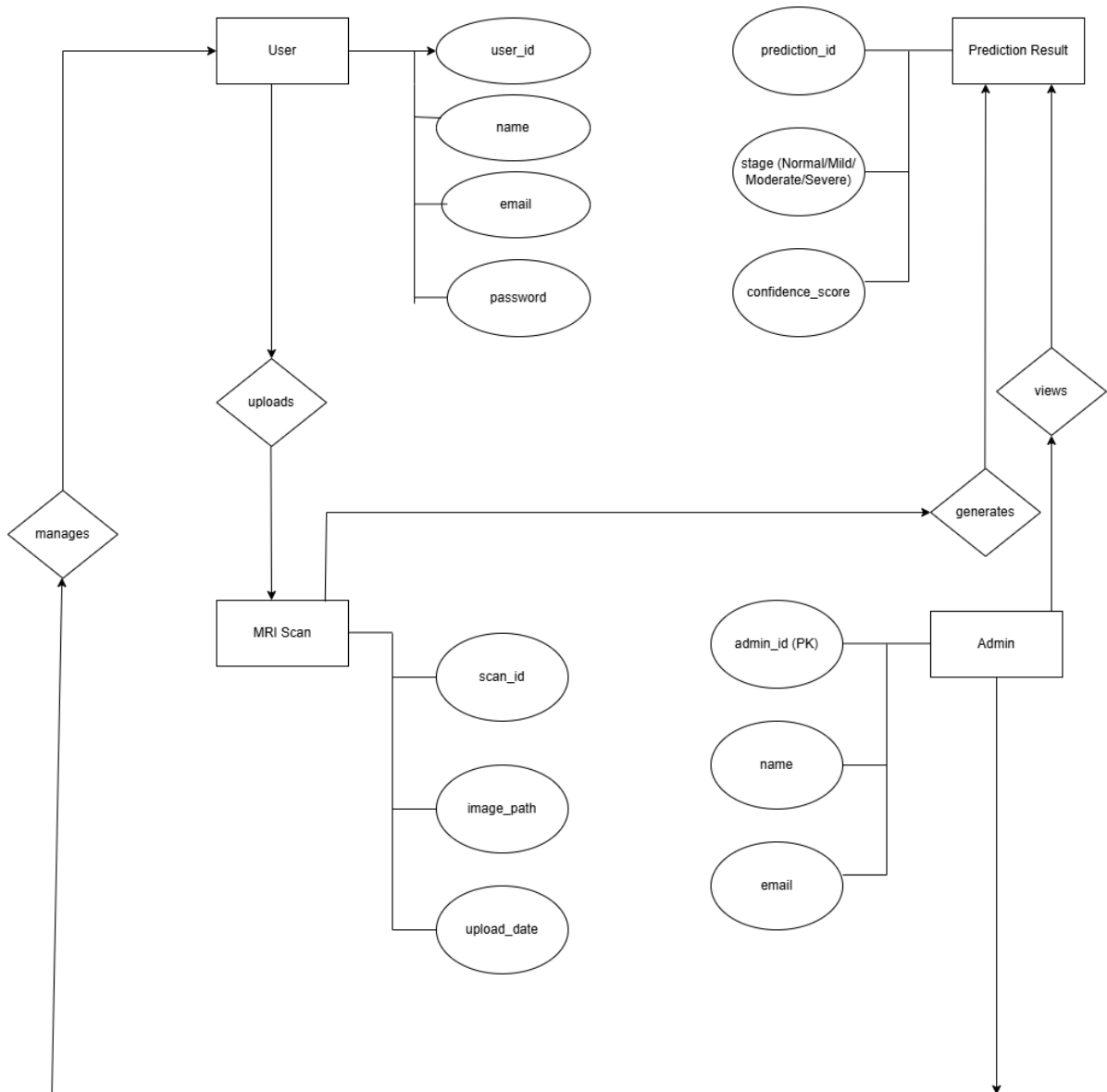


Figure 1.13: Entity Relationship Diagram of Alzheimer Prediction System

1.7.2 Database Architecture

The Alzheimers Disease Prediction System has a database architecture that's like a team effort between the client and the server. This team effort is good for managing data in a way that's safe and can handle a lot of information.

Client-Server Architecture: The system is set up so that the client, which can be a website or a mobile app sends requests to the server. The server then does what the client asks looks at the database. Sends the answers back to the client. This way of doing things makes the system more secure, faster and easier to maintain.

Cloud Storage (if we need it): We can use cloud storage to handle medical files like MRI scans. We can store these files on cloud platforms, which means they are always available can be accessed from anywhere and can handle a lot of data. This also helps the database server by not

overloading it.

Distributed Storage (for Machine Learning data): When we are working with machine learning we can use distributed storage to manage datasets. We can spread the data across computers, which helps us process the data faster do many things at the same time when training models and makes the system more reliable and able to handle problems. The Alzheimers Disease Prediction System uses this kind of storage for machine learning data to make sure it works well.

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1.8 Database Schema Diagram

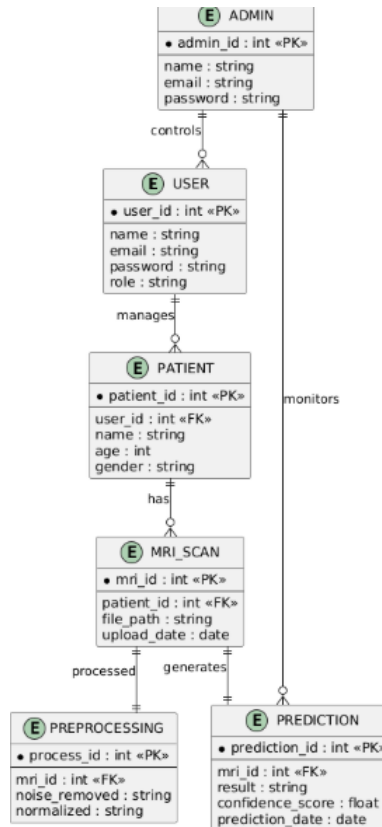


Figure 1.14: Database Schema Diagram of the Proposed System

The database schema is designed to ensure data integrity by using primary and foreign keys that maintain consistent relationships between tables. It supports efficient querying through well-structured normalization and indexed attributes. Scalability is achieved by allowing modular table expansion without affecting existing data structure. Security is enhanced by controlled access to sensitive tables and enforcing constraints that prevent unauthorized or invalid data entry.

1.8.1 Data Dictionary

Data Dictionary defines the structure of all data elements used in the Alzheimer's Disease Prediction System.

Table 1.1: Sample Data Dictionary – Patient Table

Field Name	Type	Size	Description
patient_id	INT	11	Unique identifier for each patient
name	VARCHAR	100	Patient full name
age	INT	3	Age of patient
gender	VARCHAR	10	Gender of patient (Male/Female)
mri_scan	VARCHAR	255	Path of MRI image file stored in system
prediction_result	VARCHAR	50	Alzheimer stage prediction (Normal, Mild, Moderate, Severe)
confidence_score	FLOAT	5,2	Model prediction confidence level
date_recorded	DATE	–	Date of record entry

1.9 Database Tables / Collections

This section describes the logical database design of the system. It presents the major relational tables used in the Alzheimer’s Disease Prediction System. The database structure is designed to ensure data consistency, integrity, security, and scalability.

Each table includes primary keys, relationships, and key attributes required for system functionality.

1.9.1 User Table

The User table stores authentication credentials and user profile information. It supports secure login and role-based access control within the system.

Table 1.2: User Table Structure

Field Name	Data Type	Constraint	Description
user_id	INT / UUID	Primary Key	Unique ID for each user
full_name	VARCHAR(100)	Not Null	Name of the user
email	VARCHAR(150)	Unique, Not Null	Email for login
password_hash	VARCHAR(255)	Not Null	Encrypted password
role	VARCHAR(50)	Not Null	Admin or User role
created_at	TIMESTAMP	Default CURRENT_TIMESTAMP	Account creation time
status	BOOLEAN	Default TRUE	Active/inactive account

1.9.2 Core Data Collection

This subsection describes the core data structures used in the Alzheimer’s Disease Prediction System. These collections store the primary data required for training the model, generating predictions, and managing system records.

Core Data Tables / Collections:

- **Dataset Collection** – Stores MRI image datasets used for training and testing the deep learning model.

- **Model Metadata Collection** – Stores model details such as architecture, hyperparameters, accuracy, and version history.
- **Prediction Results Collection** – Stores prediction outputs including Alzheimer’s stage, confidence score, and timestamp.
- **Patient Records Table** – Stores patient information linked with MRI scans and prediction history.

1.9.3 Relationships Between Tables

Database relationships define how different entities are connected within the Alzheimer’s Disease Prediction System.

- **One-to-One (1:1):** Each patient record is associated with one user account profile in the system.
- **One-to-Many (1:N):** One patient can have multiple MRI scans and multiple prediction records over time.
- **Many-to-One (N:1):** Multiple prediction results are generated using a single trained model version.

1.10 Entity Relationship Diagram

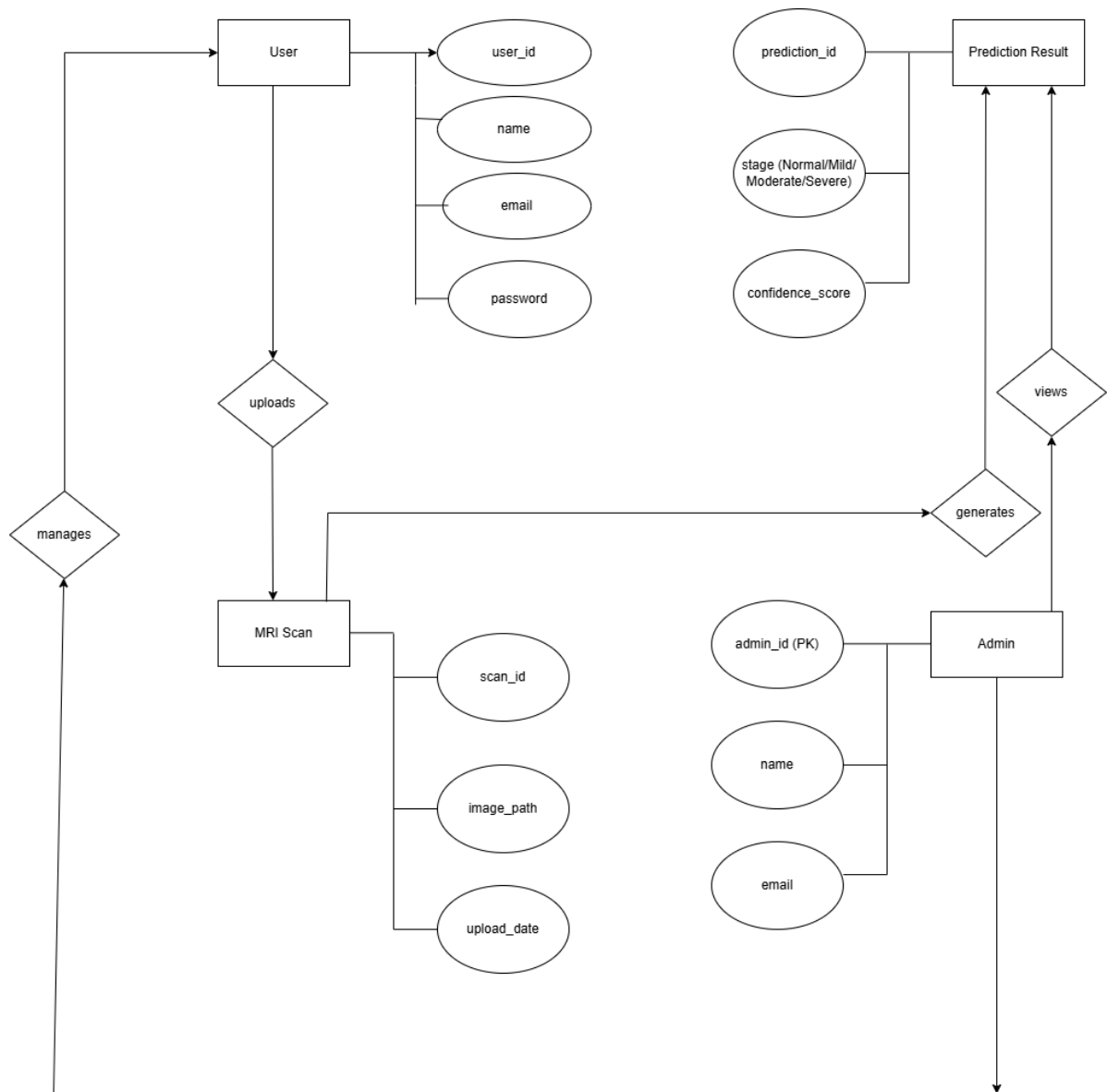


Figure 1.15: Entity Relationship Diagram

1.10.1 Security and Data Integrity

We take the security of data in the Alzheimer's Disease Prediction System very seriously. To keep data safe we do the following:

- 1 We store data and credentials securely using a technique called bcrypt.
- 2 We control who can access what in the system. There are three types of users: Admin, Doctor/Researcher and User.
- 3 We check all input data to prevent malicious data from getting into the system.
- 4 We store images like MRI scans securely and only allow access through secure sessions.

- 5 We use a communication protocol called HTTPS to protect data sent between the client and server.
- 6 We regularly back up data to prevent loss and ensure the system can recover.

1.10.2 Scalability Considerations

Our system is designed to grow with the needs of our users.

Here's how we plan to handle growth:

- 1 The system can handle MRI datasets as we get more data to train and improve our model.
- 2 We use a database structure to quickly retrieve patient records and prediction history.
- 3 Our model can be retrained with data without breaking what's already working.
- 4 We can integrate cloud-based storage for medical image datasets.
- 5 Our system is modular so we can easily add features like predicting multiple diseases.

1.11 Chapter Summary

This chapter explained the software design of the Alzheimer's Disease Prediction System. We took system requirements. Turned them into detailed plans for the architecture, behavior and data models. We defined a high-level system architecture that shows how the user interface, backend processing and machine learning model work together. We used UML diagrams to illustrate system behavior and structure. We explained the database design using entity-relationship diagrams, schema definitions and data dictionaries. This ensures that records MRI scan data and prediction results are handled properly. We included security measures like encryption, authentication and role-based access control to protect medical data. We also considered scalability to ensure the system can handle data and future improvements. Overall this chapter provides a technical foundation, for implementing the Alzheimer's Disease Prediction System.